

Assumptions:

1. Each PATIENT can schedule multiple APPOINTMENTS, while each APPOINTMENT is scheduled by one PATIENT only.
2. Each DOCTOR can handle multiple APPOINTMENTS, while each APPOINTMENT is handled by one DOCTOR only.
3. DOCTORS are classified into GENERAL_DOCTOR and SPECIALIST using total and disjoint specialization, meaning every DOCTOR belongs to exactly one subtype.
4. The Job Type attribute acts as the specialization discriminator, determining whether a DOCTOR is a GENERAL_DOCTOR or a SPECIALIST.
5. Each GENERAL_DOCTOR has the additional attribute Years_Experience.
6. Each SPECIALIST has the additional attribute Specialty_Area.
7. Each APPOINTMENT can include multiple TREATMENTS, while each TREATMENT belongs to one APPOINTMENT only.
8. A TREATMENT may require multiple MEDICINES, and a MEDICINE may be used in multiple TREATMENTS.
9. Each PATIENT can make multiple PAYMENTS, while each PAYMENT belongs to one PATIENT only.